

Anurag Bhattacharjee

London, Ontario | +1 226-998-5214 | bhattacharjee.anurag@outlook.com
<https://linkedin.com/in/anuragbhattacharjee> | <https://anuragbhattacharjee.github.io/>

PROFESSIONAL SUMMARY

Senior full-stack software engineer with 10+ years of experience building enterprise applications, REST APIs, and internal platforms. Led the initial architecture and development of a Django/React modernization of a 30-year-old CRM. Built a secure database-access platform that reduced related support requests by approximately 30%, and delivered self-service DNS management for more than 200 business customers. At Webable Digital, made architecture decisions, led production releases, worked directly with clients, and mentored junior developers. Used Claude CLI for development automation at Execulink and prototyped internal knowledge retrieval with embeddings and pgvector. Hold an M.Sc. in Computer Science with an AI specialization; my biomedical indexing research achieved 94% Precision@1.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#, Perl, Java, Groovy

Backend & APIs: FastAPI, Django, Flask, REST APIs, Node.js, Express.js, ASP.NET, ASP.NET Core Web APIs, Entity Framework, third-party integrations

Frontend: React, Angular, Vue.js, Lit, HTML5, CSS3, Bootstrap, jQuery

AI / ML: RAG, LangChain, LangGraph, embeddings, vector search, semantic retrieval, model evaluation, Hugging Face Transformers, BERT, PyTorch, TensorFlow/Keras, NLP

Data & DevOps: PostgreSQL, pgvector, MySQL, SQL Server, Supabase, Docker, Git, CI/CD, Linux, SVN

Engineering: System design, data pipelines, authentication, session management, RBAC

EXPERIENCE

Full Stack Software Engineer, Execulink Telecom, Woodstock, Ontario

Aug 2023 – Jun 2026

- Led the initial architecture and development of Viridian, a Django/React modernization initiative for a 30-year-old Perl CRM; built its modular backend foundation, protected routes, reusable components, and integration path for future API-driven services.
- Delivered a self-service DNS management platform using Perl, Lit, and the PowerDNS REST API, enabling internal teams and more than 200 business customers to manage their DNS records directly.
- Prototyped RAG-based internal knowledge retrieval in Python using embeddings, pgvector/Supabase, and LLM-based semantic search, with a focus on ingestion and retrieval quality.
- Designed and built a secure database-access platform end to end, including the frontend, backend, guided workflows, role-based access controls, and audit logging. Enabled support staff to view and update records without writing SQL, reducing related support requests by approximately 30%.
- Automated ONT inventory, assignment, snapshots, and status checks in fibre-order workflows; cut manual handling by approximately 70% and reduced errors for support and provisioning teams.
- Built tools for IP address management, service monitoring, inventory alerts, usage reporting, and telecom status dashboards, translating complex SQL and business rules into practical workflows for internal teams.

Senior Developer, Webable Digital, Bangladesh

Sep 2018 – Aug 2021

- Represented the development team in client meetings, led architecture decisions and production releases, and mentored junior developers on design and code quality.
- Built a customer support chatbot from scratch using Python, Flask, MySQL, NLP, and REST APIs.
- Integrated the chatbot with the commerce platform's Facebook Messenger and Instagram channels, routing customer messages to its API and returning generated replies in the same conversation.
- Created several datasets to support intent-classification accuracy.
- Gathered requirements directly from clients, translated them into technical solutions, and led deployments from development through production release.
- Helped build a messenger-based commerce experience that allowed customers to browse products and interact with businesses directly through chat.
- Designed customer showcase websites with PHP/Laravel, Vue.js, React, JavaScript, HTML, and CSS.

Software Engineer, IQVIA, Bangladesh

Jul 2015 – Aug 2018

- Built enterprise KPI dashboards with Angular, TypeScript, ASP.NET, C#, and SQL to help healthcare clients visualize business performance and make data-driven decisions.
- Built ASP.NET Core Web API endpoints in C# and worked with Entity Framework and SQL Server.
- Refined drug-information processing and search logic, reducing retrieval time to approximately one-sixth of its previous duration.
- Contributed to an in-house C# framework built on ASP.NET that standardized reusable dashboard components and reduced development time by approximately 60%.
- Received the Client Delight Award in 2017 and won the company hackathon the same year.

- Contributed to sbiCloud, an enterprise resource planning platform used to manage employee and operational data across BRAC Bank's international operations, working with Groovy, Grails, and PostgreSQL.
- Gained early experience maintaining a large corporate software system, collaborating within an engineering team, and understanding how data, business processes, and software development come together in an enterprise environment.

PROJECTS & OPEN SOURCE (<https://github.com/anuragbhattacharjee>)

AI Retrieval Pipeline | Python, FastAPI, PostgreSQL/pgvector, Embeddings 2026

- Ongoing RAG application covering document ingestion, chunking, embedding generation, semantic retrieval, and evaluation, with a modular FastAPI backend for comparing retrieval strategies and measuring answer relevance.

My Portfolio Redesign | React, TypeScript, CI/CD 2026

- Redesigned my Portfolio, implemented with React and TypeScript. Deployed as my website in git pages, built CI/CD pipelines for easy deployment.

Clockin | Python, Linux, Snap 2026

- Built and published a lightweight command-line time tracker for Linux with tagged work sessions, weekly and monthly summaries, local JSON storage, and CSV export. Designed it to run without accounts, cloud services, or external Python dependencies.

Discord NLP Bot | Python, PyTorch, Discord API 2020

- Built and deployed a real-time Discord bot that analyzes the sentiment, emotion, and sarcasm of live messages using a deep-learning model and responds with a contextually relevant emoji. The open-source project has received 19 GitHub stars and three forks.

LEADERSHIP & TEACHING EXPERIENCE

Board Member, Community of Guardians, London, Ontario Dec 2023 – Present

- Serve as a registered board member, contributing to organizational planning, governance, and long-term decision-making.
- Review organizational proposals and presentations, contribute to board discussions, and vote on key decisions.

Volunteer Chief Technology Officer, Community of Guardians, London, Ontario Jul 2023 – Aug 2023

- Led early technology planning and website development, helping establish the organization's initial technical direction.

Teaching Assistant, Western University, Department of Computer Science, London, Ontario Sep 2021 – Dec 2022

- Supported more than 200 students across four undergraduate computer science courses covering JavaScript programming, information systems, software development tools, and foundational computer science.
- Led tutorials, reviewed assignments, explained technical concepts, and helped students improve their programming and problem-solving skills.
- Courses: Foundations of Computer Science; Information Systems and Design; Programming Modern Survival Skills II; Approachable Apps—A Gentle Introduction to Programming Using JavaScript.

RESEARCH

BERT-KenMeSH, Western University 2023

Developed a deep-learning model for automatic MeSH indexing of biomedical articles, combining BERT with the KenMeSH approach to improve how PubMed research is categorized and retrieved. Achieved 94% Precision@1 across more than 35 million MEDLINE articles.

Supervisor: Dr. Robert E. Mercer

Thesis: <https://uwo.scholaris.ca/items/5f8d2433-f707-4fdc-9049-316a8dfb6db6>

Bangla Speech Corpus for Large-Vocabulary ASR, KUET 2014

Built a Bengali dialect speech corpus using recordings and utterances from 15 speakers to support automatic speech recognition.

Supervisor: Dr. Pintu Chandra Shill

EDUCATION

- **M.Sc. Computer Science, AI Specialization (Thesis).** GPA 3.9/4.0, Western University 2021 – 2023
- **B.Sc. Computer Science & Engineering (Thesis).** KUET, Bangladesh 2010 – 2014

CERTIFICATIONS & RECOGNITION

- Ivey Business School Innovation Scholar 2023
- Vector Institute AI Specialization 2023